Sujit Kumar Kamaraj

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EDUCATION

University of Utah **Aug 2023 - Present**
PhD, Computer Science Salt Lake City, UT
Masters of Science, Computer Science – 3.98 CGPA

Vellore Institute of Technology **Aug 2019 - May 2023**
BTech, Computer Science and Engineering – 8.77/10 CGPA Vellore, Tamil Nadu, India

PUBLICATIONS

1. **Kamaraj, S. K.**, & Pandey, V. (2026). *Crossroads: A Self-Administered Digital Cognitive Assessment for Older Adults via Fine-Finger Tracking on Touchscreens*. In Proceedings of the International ACM SIGACCESS Conference on Computers and Accessibility.

2. **Kamaraj, S.K.**, & Pandey, V. (2026). *Documentation Traps in Classroom Settings*. "Nobody Did This": Contribution, Originality, and Accountability in Agent-Mediated Collaboration - Workshop at the 2026 ACM Conference on Computer-Supported Cooperative Work and Social Computing.

3. **Kamaraj, S. K**., & Pandey, V. (2027). *Investigating the Process-Platform Gap: How a Patient Community's Efforts Teach us About the Limits of Social Platforms in Supporting Institutional Processes*. Proceedings of the ACM on Human-Computer Interaction, 10(1), 1-25.

4. McNutt, A., He, S., **Kamaraj, S. K**., Bambroo, P., Jadidi, N., Bovard, J., & Han, C. (2025). *Teaching Critical Visualization: A Field Report*. IEEE VIS Workshop on Visualization Education, Literacy, and Activities (EduVIS) 2025. arXiv preprint arXiv:2508.02592.

5. **Kumar, S.**, Rajesh, D. D., Pranesh, S., Kollipara, V. H., Agrawal, G. K., & Anbarasi, M. (2022). *Classification of Indian media titles using deep learning techniques*. International Journal of Cognitive Computing in Engineering, 3, 114-123.

In Submission

6. Pandey, V., **Kamaraj, S. K.**, Jones, G.M., Satyanarayan, A. (2026). *An Ecosystem of Evidence: How Communities Complement, Contest, and Contextualize Institutional Ways of Knowing*. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.

Posters/Talks

7. **Sujit Kumar Kamaraj**, Vineet Pandey. *Tracking the Learning Process in Student Work to Make Human-AI Collaboration Visible*. One-U RAI Annual Symposium, Salt Lake City, UT.

8. **Sujit Kumar Kamaraj**, Vineet Pandey. *Tracking the Learning Process in Student Work to Make Human-AI Collaboration Visible*. Utah AI Convergence 2026, Salt Lake City, UT.

9. **Sujit Kumar Kamaraj,** Vineet Pandey. *Teaching with Machines: How Instructors Are Navigating Opportunities and Challenges of Generative AI*. 10th Digital Humanities Utah Symposium (DHU10) 2026**,** Ogden, UT.

10. **Sujit Kumar Kamaraj,** Vineet Pandey. *Teaching with Machines: How Instructors Are Navigating Opportunities and Challenges of Generative AI*. AI Alliance Academic Educator Work Group 2026, Virtual.

11. Nastaran Jadidi, Carly Atwell, Asma Khan, **Sujit K. Kamaraj,** Vineet Pandey. *One Size Does Not Fit All: Two Approaches to Better Support Women in Understanding and Managing their Health*. Women's Health Symposium 2025, University of Utah, Salt Lake City, UT (**First Prize in Posters**)

12. **Sujit Kumar Kamaraj**, Vineet Pandey. *Cognitive Profiles Using Fine-Finger Performance on a Web-*

Based Task for Touchscreens. Conference for Advancing Participatory Sciences 2025, Portland, OR.

13. **Sujit Kumar Kamaraj**, Vineet Pandey. *Cognitive Profiles Using Fine-finger Performance on a Web-based Task for Touchscreens*. DELPHI data science initiative, November 2024, Salt Lake City, UT.

HONOURS & AWARDS

1. One-U Responsible AI Initiative Graduate Student Fellow at the University of Utah (2026 -)

WORK EXPERIENCE

The University of Utah – Research Assistant **Oct 2023 – Present**
Human Computer Interaction Researcher Salt Lake City, Utah, US

- Currently building AI tools that track the learning process passively–without interrupting students' workflow–while giving instructors direct, observable evidence of AI use.
- Conducted a multi-site interview study with instructors on how generative AI is transforming design education and instructional practices.
- Developed touchscreen-based cognitive assessment and tested it with older adults and students.
- Analyzed how rare-disease communities use social platforms to participate in institutional processes.

The Centre of Excellence for Road Safety (CoERS) – IIT Madras **Dec 2022 – Jun 2023**
Computer Vision Research Intern Chennai, Tamil Nadu, India

- Spearheaded the development of cutting-edge computer vision solutions aimed at enhancing road safety across Indian roadways.
- Trained and evaluated the YOLOv7 deep learning model, achieving a F1 score of 61.2% on an extensive dataset comprising 10,000 images and encompassing 34 diverse classes such as cars, buses, bikes, traffic signs, and pedestrians.
- Implemented innovative techniques utilizing image contours to enhance detection in nighttime and low-light conditions.
- Integrated the CRAFT and PARSeq models seamlessly to establish a high-performance real-time scene text recognition pipeline.

University of Ottawa – Mitacs Research Internship **May 2022 – Aug 2022**
Research Intern Ottawa, Canada

- The goal of the project was to study existing feature selection methods and come up with a novel feature selection algorithm that addresses the class-imbalanced problem in high-dimensional datasets.
- Implemented and tested standard feature selection methods like ReliefF, Chi-Square and SVM-RFE on high-dimensional class-imbalanced datasets.
- Implemented a novel feature selection method that was developed for high-dimensional class-imbalanced datasets.
- Initial results show that the novel feature selection algorithm developed performs better than existing methods.

Vegam Industries **May 2021 – Sep 2021**
NLP Intern Vellore, Tamil Nadu, India

- Worked on text generation and paraphrasing of text based on the given situation and emotion.
- Worked on text generation models like GPT-2, BERT, LSTMs and Senti-GAN.
- T5 and BART transformer models were used for the paraphrasing task.

Samsung R & D Institute **Oct 2020 – Mar 2021**
Project Intern Bengaluru, Karnataka, India

- Engineered a language classifier for media titles (movies and songs) across six languages, optimizing text-to-speech pronunciations.
- Implemented NLP and machine learning techniques, including N-Gram, SVM, Neural Networks, MuRIL, LSTMs, and tokenization, achieving a peak accuracy of 92%.

REVIEWING

1. Associate Chair for Poster Track - ACM Conference on Human Factors in Computing Systems, CHI 2026

TEACHING ASSISTANT

CS 3540 - Designing Human-Centered Systems Kahlert School of Computing, University of Utah
Fall 2025 (N=150 students); Fall 2024 (N=136 students)

CS 6635 – Visualization for Scientific Data Kahlert School of Computing, University of Utah
Spring 2025 (N=52 students)

OUTREACH ACTIVITY

Juvenile Care VIT **Nov 2019 – May 2023**
Senior Core Member Vellore, Tamil Nadu, India

- Juvenile Care VIT is a youth led organization working for child rights and child abuse eradication.
- We provided a platform for youth to come forward and work for the alleviation of society.
- I was part of the "Design Team" and created multiple posters for the social media page.